

Interface Bonding in Linux



Ever since I started playing around with Linux and started building my own routers for a stable and fault-tolerant network, I've longed for a technology that could offer higher data rates by combining two or more network links into one single device. I found my answer in Interface Bonding.

Bonding is a technique that combines two network interface cards into one logical device, multiplying your bandwidth. Before we actually get into bonding and implementing it on Linux, let's first step back and understand link aggregation.

Link aggregation, as per IEEE 802.1AX-2008, is used to increase the speed of a link by bundling network cables/ports in parallel. There are a lot of other terms for link aggregation, including Ethernet bonding, NIC teaming, link bundling, and trunking. By achieving link aggregation, we can address two main problems: bandwidth limitations, and lack of redundancy. The main protocol used to achieve bonding is called LACP (Link Aggregation Control Protocol). It allows a bundled device to negotiate an automatic bundling of links by sending LACP packets to the peer. Figure 1 visualises a bonded/teamed link from a server to an Ethernet switch.

Getting started

Almost all current Linux distributions should already have the bonding driver installed. Check for it with the following code:

```
[root@bastion ~]# modprobe -l | grep bonding
kernel/drivers/net/bonding/bonding.ko
```

If you see the module listed, you don't need to do anything more here. If your distro does not have a driver installed, then you'll need to compile bonding from source. Compiling the kernel is out of scope for this article, so please refer to the many available how-tos or other references. The main points to remember are that you should select 'Bonding driver support' in the 'Network Device Support' section, and ensure you configure it as a module, since having it as a module is currently the only way to pass parameters to the driver, or to configure more than one bonding device.

There are two ways to configure bonding, and we'll cover each of them in turn:

- Passing options to the bonding driver module at load time or with init scripts
- Or, if administration tools on your system don't support those, then by manual configuration.

Passing bonding driver options as parameters

Create an `ifcfg-bond0` configuration file under `/etc/sysconfig/network-scripts` (you can copy the existing `ifcfg-ethX` file to `ifcfg-bond0` and edit that) as follows (be sure to change the lines `IPADDR`, `NETMASK`, `NETWORK` and `BROADCAST` to match your network configuration):

```
[root@bastion network-scripts]# cat ifcfg-bond0
DEVICE=bond0
```